Advance Python Assignment

Topic-9: SQLite3 and PyMySQL (Database Connectors)

Q1. Introduction to SQLite3 and PyMySQL for database connectivity.

Ans.
SQLite3 and PyMySQL are database connectors used in Python to interact with databases. SQLite3 is a lightweight, built-in database that does not require a separate server and is easy to use for small applications. PyMySQL is used to connect Python with MySQL databases and is suitable for larger applications. These connectors allow us to store, retrieve, and manage data using SQL queries. They are widely used in web development and data management. Using these tools, Python programs can easily work with databases.

Example:
```
import sqlite3

conn = sqlite3.connect("mydb.db")
print("Database connected")
conn.close()
```

Q2. Creating and executing SQL queries using SQLite3 in Python.

Ans.
In Python, we can use SQLite3 to create and execute SQL queries. After connecting to the database, we use a cursor object to run SQL commands like creating tables, inserting data, and fetching records. SQL queries are written as strings and executed using the execute() method. This allows Python programs to manage data easily. It is very useful for storing and retrieving structured data in applications.

Example:
```
import sqlite3

conn = sqlite3.connect("mydb.db")
cursor = conn.cursor()

cursor.execute("CREATE TABLE IF NOT EXISTS student(id INTEGER, name TEXT)")
```

```
cursor.execute("INSERT INTO student VALUES(1, 'Anshu')")

conn.commit()
conn.close()
```